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A FET analogue switch for pulsed NMR receivers

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Abstract A low-noise FET analogue switch is described which has been used to deal with some of the problems encountered in optimizing the design of a probe for a pulsed NMR spectrometer intended for work on solids. The switch circuit has been used in association with two spectrometers, one for microsecond RF pulses and the other for RF pulsewidths of up to 1 s. In each case the resolution time achieved was limited by the transmitter coil ring down. The switch has negligible insertion loss when on and 84 dB attenuation at 15 MHz when off, thus affording good protection of the receiver system.

1 Introduction

In the design of a pulsed NMR spectrometer for work on solids the requirements of the transmitter and receiver coils are in conflict. Before the nuclear signal following an RF pulse can be observed, the ringing of the probe coils must be reduced to a level much less than that of the nuclear signal induced in the receiver coil. Spin-spin relaxation times of nuclei in solids may be as short as a few microseconds, so the transmitter coil must have a low Q -factor to give a short ringing time. On the other hand, the receiver coil should have a high Q -factor to maximize the nuclear signal-to-noise ratio. An orthogonal coil configuration allows these conflicting requirements to be satisfied at least to some degree.

Even with the crossed-coil attenuations of 60 dB or more which are readily obtainable, the RF voltage induced in the receiver coil during the intense RF pulses at the transmitter coil is typically a few volts. So the ringing of the receiver coil will also contribute to the resolution time of the spectrometer. Crossed diodes will clamp the voltage induced in the receiver coil to a few tenths of a volt, and this affords some decrease in the resolution time. Another consideration is that the later stages of a NMR receiver will be saturated by such an input or even paralysed if the pulses are very long; the recovery from saturation may add further to the resolution time. In experiments using very long pulses or sequences of pulses, the gain stability may be degraded owing to overheating of passive as well as active elements in the amplifiers.

We wish to describe an FET analogue switch which we have used to mitigate these problems. The analogue switch is

connected between the tuned receiver coil and the signal preamplifier, and has been designed around the following criteria:

- (i) high isolation between receiver coil and signal preamplifier during the RF pulses;
- (ii) reduction of receiver coil Q -factor during the RF pulses;
- (iii) minimal introduction of transients into the signal path when the device is switched on and off.

2 The analogue switch

The simplest FET analogue switch is the circuit of figure 1. It is on when the gate voltage is zero, assuming that the input voltage V_i is very small, and is off when the gate voltage is greater than that required for pinch-off. This circuit provides

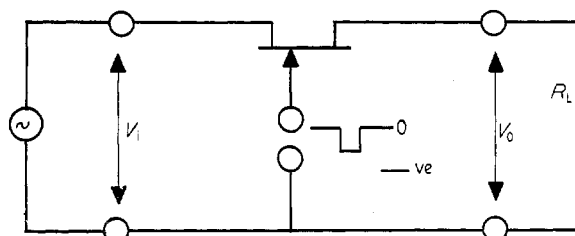


Figure 1 A simple FET analogue switch

considerable isolation between input and output, but relatively large transient currents flow into the signal path. The size of these transients increases with the gate-drain capacitance, the rate at which the gate bias voltage is changed and the amount by which the negative bias voltage exceeds the pinch-off value.† With $R_L = 10 \text{ k}\Omega$, and a Siliconix FET type VCR 2N, transients several tenths of a volt in amplitude were observed across R_L when $\sim 4 \text{ V}$ pulses of 400 ns approximate risetime were applied to the gate. Such transients easily saturated a NMR receiver.

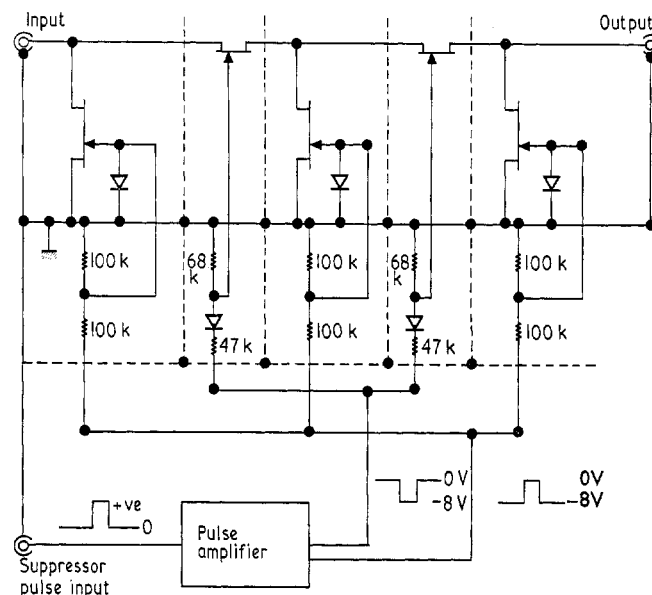


Figure 2 A useful pi-section analogue switch. All FETs are Siliconix VCR 2N and all diodes are OA47; the switch is connected between the tuned receiver coil, shunted by the usual crossed pair of silicon diodes, and the receiver preamplifier

† *An Introduction to Field Effect Transistors* J Watson, Siliconix Ltd, Swansea; *Shunt and Series, FET Choppers* P Conaway and L Delhom, Texas Instruments, Bedford, England

A more effective circuit has two pi-sections of n-channel FETs; the series and shunt devices are driven by complementary pulses. Greater isolation is achieved in the off state, and the transient voltages produced at the output are about two orders of magnitude smaller than with the simple circuit. This reduction occurs because the transients originating in the series and shunt FETs are of opposite sign and approximately cancel. In addition, in the off state the switch shunts the input with a low resistance.

Figure 2 shows the circuit of an FET analogue switch that has been used in this laboratory. The Siliconix VCR 2N FETs were chosen for their linearity at low signal levels. The pulse amplifier comprised two common emitter stages, the gate drive pulses being taken from the two collectors. There is no reason why TTL compatible circuitry should not be used.

3 Performance

The circuit of figure 2 has been used to protect the receiver in pulsed nuclear double resonance experiments (Mansfield and Grannell 1972, Grannell 1972, Grannell *et al.* to be published). In these experiments two intense RF fields were used with pulsewidths up to 1 s. The analogue switch was pulsed off for the duration of the RF pulses, and for about 10 μ s afterwards so that the receiver remained protected and the receiver coil was kept at a low Q -value during the ring down of the transmitter coil. The 15 MHz receiver circuit was essentially that of Mansfield and Powles (1963) and had a bandwidth of 450 kHz. Orthogonal transmitter and receiver coils were used with a crossed-coil attenuation (defined as the ratio of the voltage applied across the transmitter coil to that measured across the open-circuit receiver coil) measured as 64 dB.

In the on state the analogue switch did not perceptibly degrade the receiver linearity, which was better than 2%, for signals from the threshold of detection up to at least 250 μ V. It contributed negligible noise and had a negligible insertion loss when loaded by a 10 k Ω resistance. The input capacitance was found to be 25 pF. The attenuation in the off state was measured as 84 dB at 15 MHz. This value is approximately consistent with the attenuation being limited by the drain-source resistance of the shunt FETs, nominally 20–60 Ω , and the gate-source and gate-drain capacitances of the series devices, nominally 7.5 pF each. The rise- and fall-times of

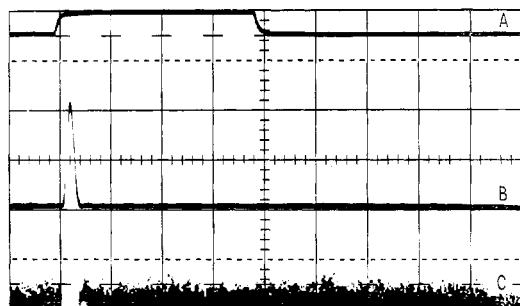


Figure 3 Transients observed at the receiver output when the state of the analogue switch is changed. Time scale for all the traces is 5 μ s per large division. A, the suppression pulse applied to the analogue switch; B, the receiver output: a transient associated with the on-off transition is clearly seen, but no such transient at the off-on transition appears; vertical scale: 1 V per large division; C, the receiver output: further amplification shows that the amplitude of any off-on transient is below noise level; vertical scale: 50 mV per large division

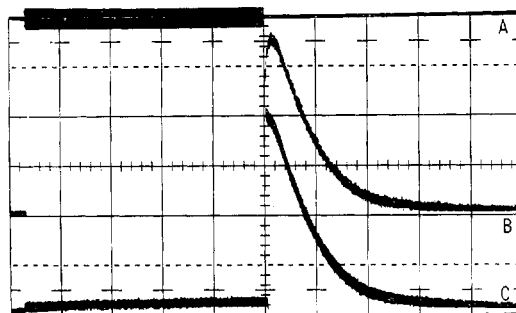
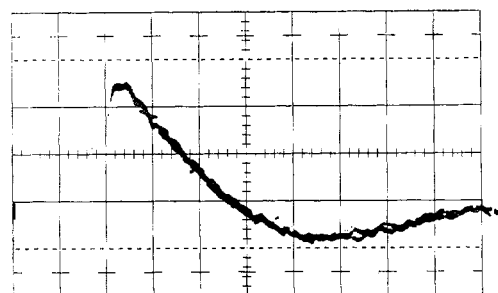


Figure 4 A, 4.8 ms spin locking RF pulse applied to protons resonant at 15 MHz in a water sample; vertical scale, 10 kV per large division; B, the proton free decay following the spin locking pulse. The receiver is overloaded during the RF pulse and the decay shape is distorted for about 200 μ s following recovery from the RF pulse. Vertical scale: 1 V per large division; C, the proton free decay following the spin locking pulse. Here the FET analogue switch protects the receiver during the RF pulse and the output remains nearly zero. The decay shape is now undistorted. Vertical scale: 1 V per large division. Time scale for all traces: 1 ms per large division

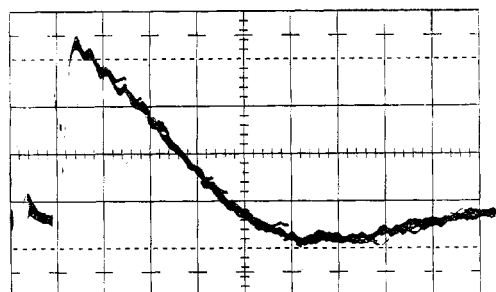
the gate drive pulses from the amplifier in figure 2 were about 400 ns. The transients produced at the output were 20 mV (on-off), and 1 mV (off-on). It was found that the off-on transient could be reduced at the expense of the on-off transient by switching the series devices 300 ns or so before the shunt ones. This delay occurred fortuitously in the pulse amplifier circuit already described. The transients could be reduced further at the expense of a loss of attenuation by trimming the pulse at the gate of each FET to the measured pinch-off voltage of that FET, nominally 3.5–7.0 V. Figure 3 shows the transients induced in the receiver system when the state of the analogue switch is changed. Although there is a large transient induced in the receiver at the on to off transition, no transient associated with the off to on transition is evident.

The effectiveness of the FET analogue switch is demonstrated by figure 4. Protons in a water sample were spin locked for 4.8 ms and, when the switch was not turned off during the RF pulse, the free decay following the pulse is clearly distorted. In a similar experiment where protons in adamantane $C_{10}H_{16}$ were spin locked for 100 ms, the free decay suffered a 30% apparent loss in amplitude and some shape distortion when the receiver was not protected. The small amount of break-through (figure 4, C) during the 1 kV RF pulse was found to have been caused by RF leakage on to the HT line of the receiver. Good shielding and the absence of ground loops were essential. The resolution time of the double resonance spectrometer was not improved significantly by the analogue switch because transmitter coil ring down was dominant; the transmitter coil had a Q -factor of about 25 compared with about 15 for that of the receiver coil.

The circuit of figure 2 has also been used in a 9 MHz multiple-pulse spectrometer for line-narrowing experiments in solids, where short resolution times are crucial (Mansfield *et al.* 1973). The probe and receiver of this spectrometer were similar to those outlined for the 15 MHz double resonance spectrometer. A useful decrease in resolution time from 12 to 7 μ s was obtained, in addition to protection of the receiver from overload during the RF pulses. Here, we define the resolution time as the interval between the start of the



(a)



(b)

Figure 5 The receiver output during and after a $1 \mu\text{s}$ 90° RF pulse on ^{19}F in CaF_2 , $H_0 \parallel [100]$, resonant at 9 MHz.

(a) The receiver is saturated during the RF pulse and the resolution time is $12 \mu\text{s}$. (b) Here the analogue switch prevents saturation by the RF pulse, and the resolution time is reduced to $7 \mu\text{s}$. Vertical scales: 0.05 V per large division, horizontal scales: $5 \mu\text{s}$ per large division

$1 \mu\text{s}$ RF pulse and the time at which the receiver output is within 5% of the nuclear signal amplitude for any signal phase. The improvement in resolution time is illustrated by figure 5. In this case the transmitter and receiver coils had Q values of about 10 and 20 respectively.

4 Conclusion

A pi-section FET analogue switch may be deployed in a pulsed NMR spectrometer to prevent saturation and paralysis of the receiver and to reduce the contribution of the receiver coil to the total resolution time. The gain stability of the receiver during long RF pulses is enhanced and the receiver coil may have a high Q -factor for good signal-to-noise ratios. Although we have discussed the analogue switch in the context of an orthogonal-coil probe only, it should also prove useful in the case of single-coil probes such as that described by Clark and McNeil (1973).

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